

Business Management Systems

Supply Chain Management

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We respectfully acknowledge the Kaurna, Boandik, and Barngarla First Nations Peoples and their Elders past and present, who are the Traditional Owners of the lands that are home to our campuses across Adelaide and South Australia.

Outline

- What is supply chain
- Complex supply chains
- Supply chain processes
- Automation of information flow

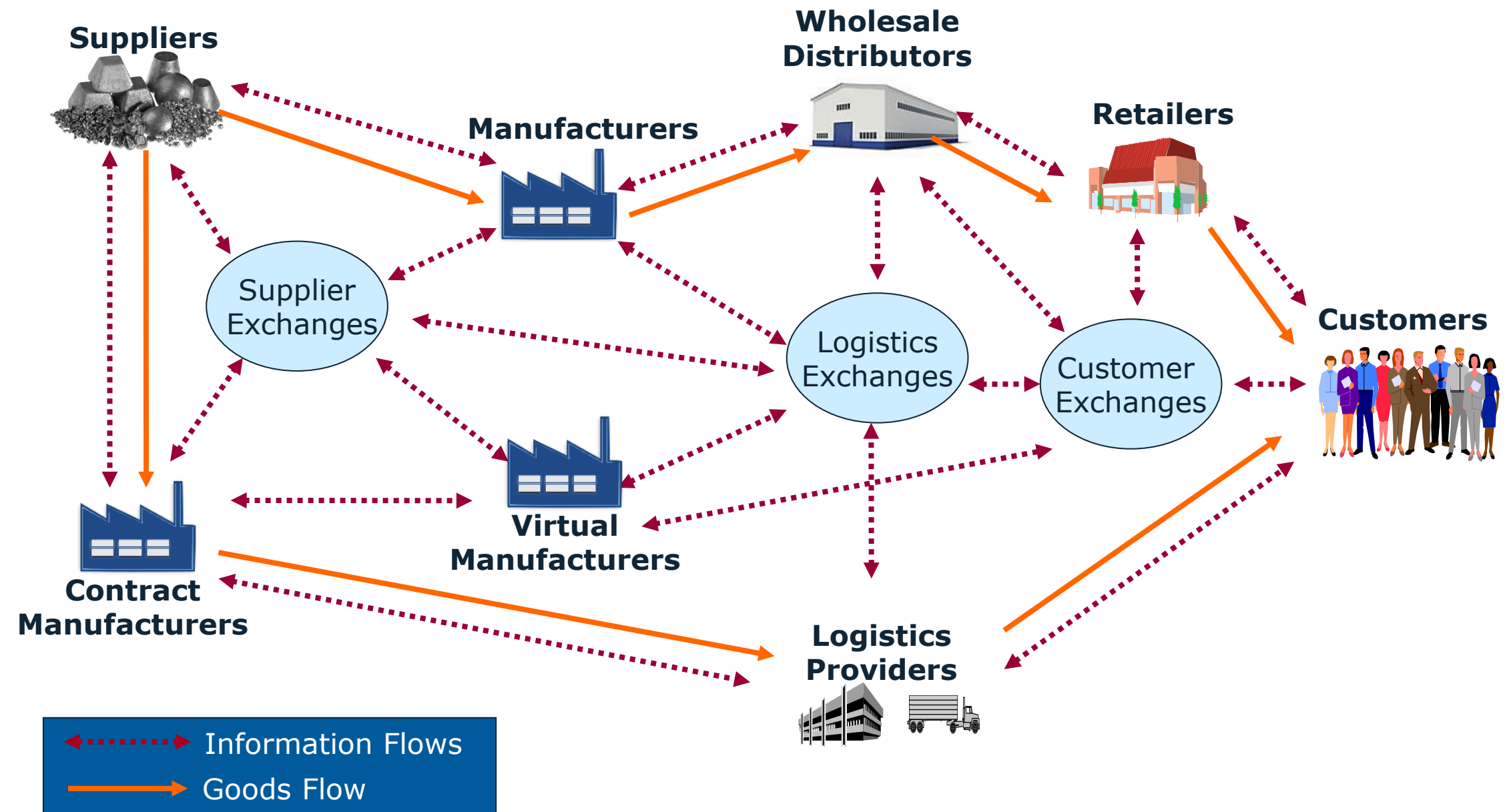
What is a Supply Chain?

- A supply chain is a network of facilities and distribution options that performs the functions of procurement of materials, transformation of these materials into intermediate and finished products, and the distribution of these finished products to customers.
- Traditional supply chain:

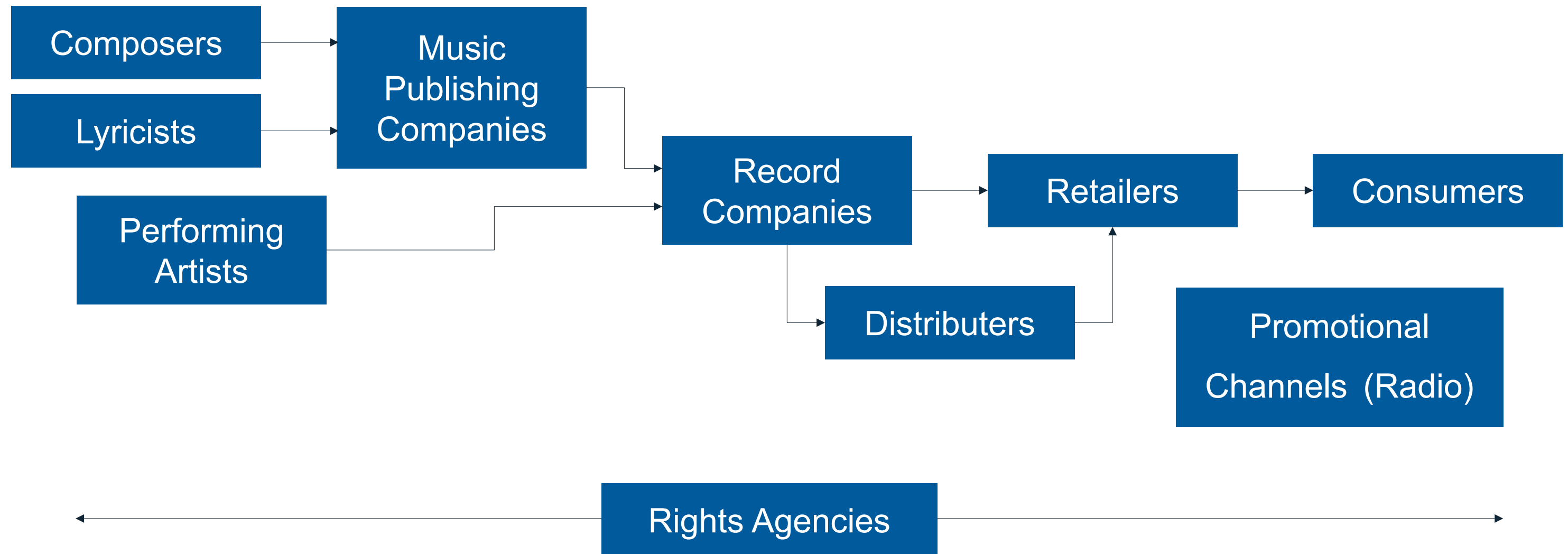


An Internet-Connected Supply Web

- A more complex picture with parallel supply lines and multiple manufacturers is what usually exist in both traditional and e-business enabled supply chains.

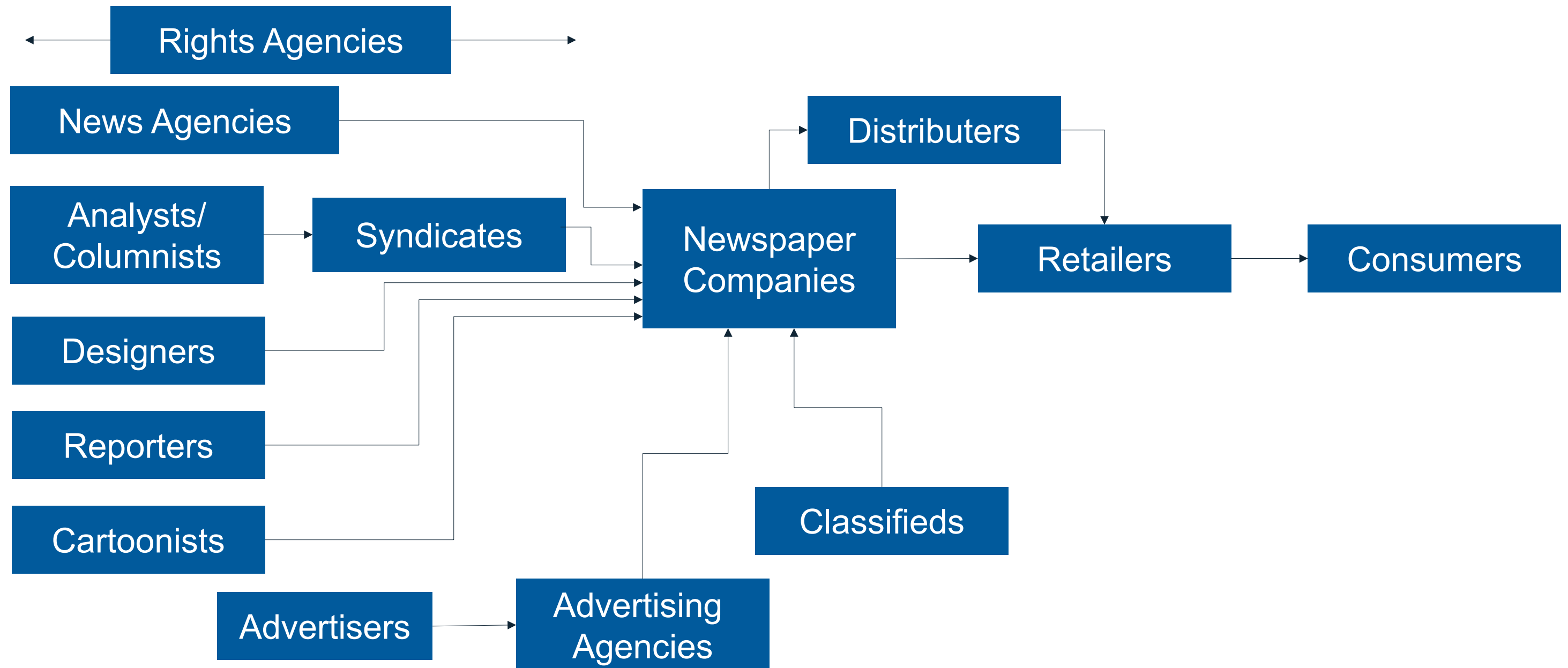


Industry Structure (Music – pre-Online)



Source: HBPS

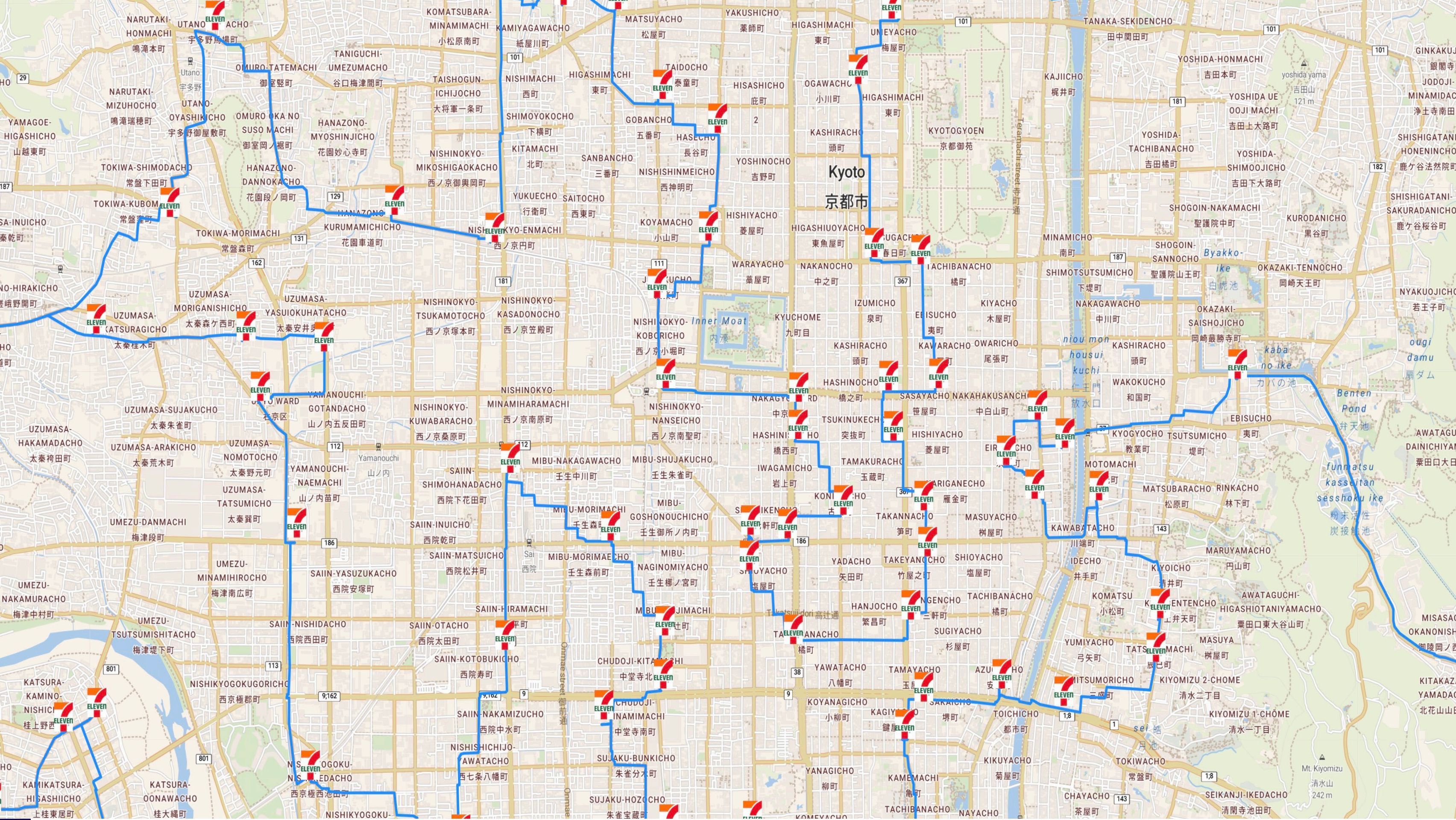
Industry Structure (Newspaper – pre-Online)



Essential to Business Success

- Examples of supply chains and process automation
 - 7-Eleven: ~20,000 stores in Japan





7-Eleven Delivery Trucks

- Goods get continuously delivered to stores in trucks



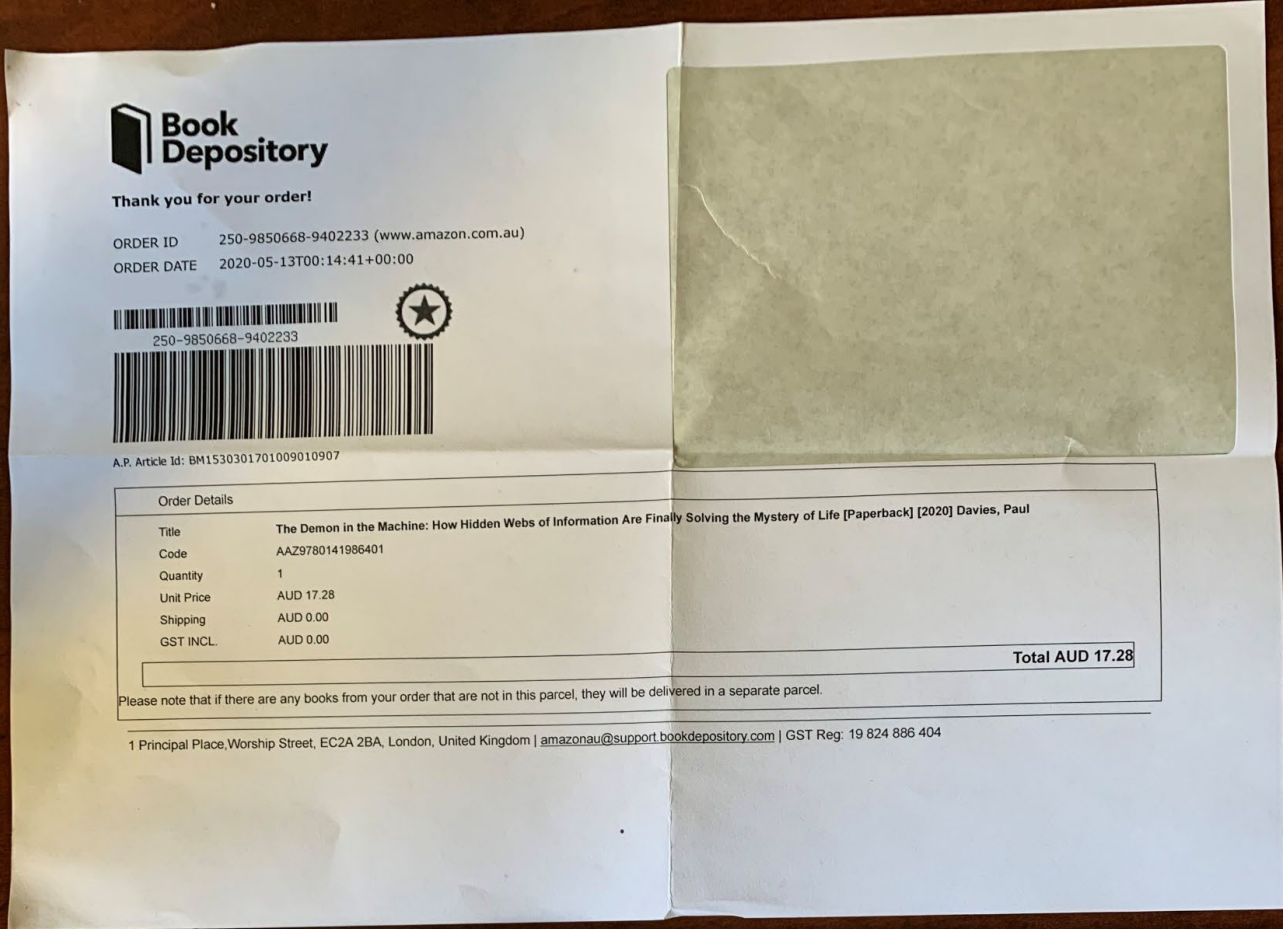
Essential to Business Success

- Amazon: Connecting many suppliers to many customers
- Streamlining delivery process with several entities
- Their own delivery network



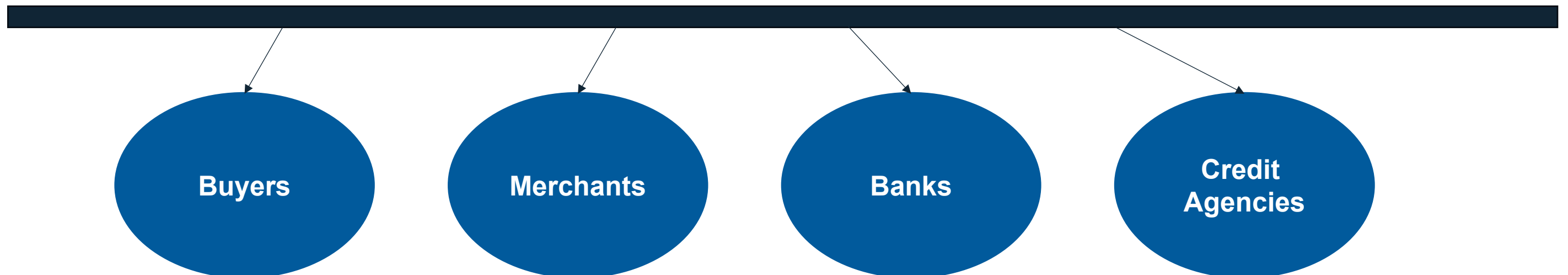
Book Delivery from the UK

- A personal example:



Platform Businesses

- Some businesses create a platform through which information flows between different entities
- Many such businesses exist (and have existed for a long time)
 - Example: Visa card platform



Amazon Platform

- Amazon can also be considered as a platform
- Connecting merchants, buyers and logistics companies together
 - In these cases, no warehousing at Amazon is needed



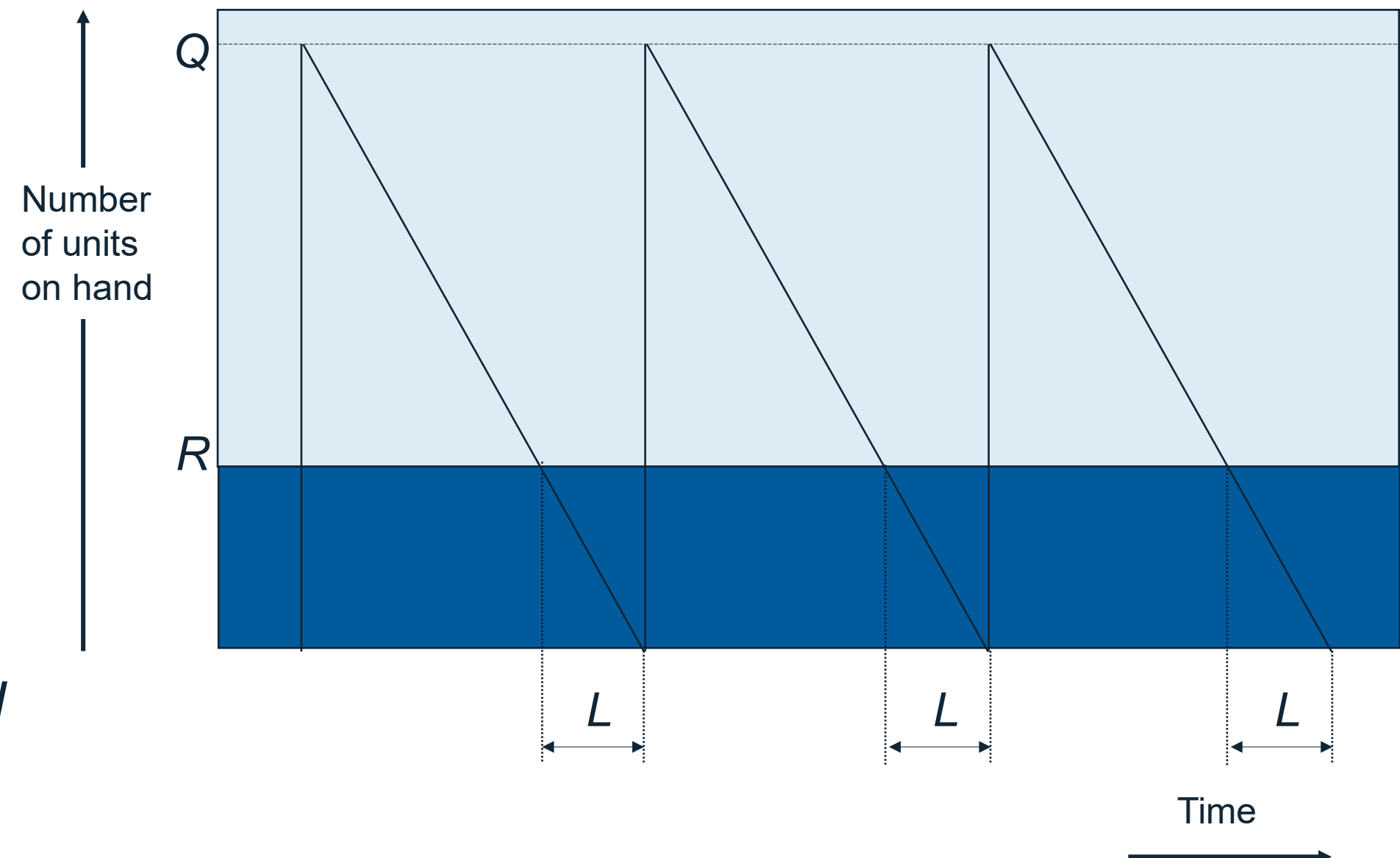
More Supply Chain Examples

- Other examples of supply chains:
 - Dell: Build to order
 - Alcoa: Resetting the aluminium manufacturing process in response to demand



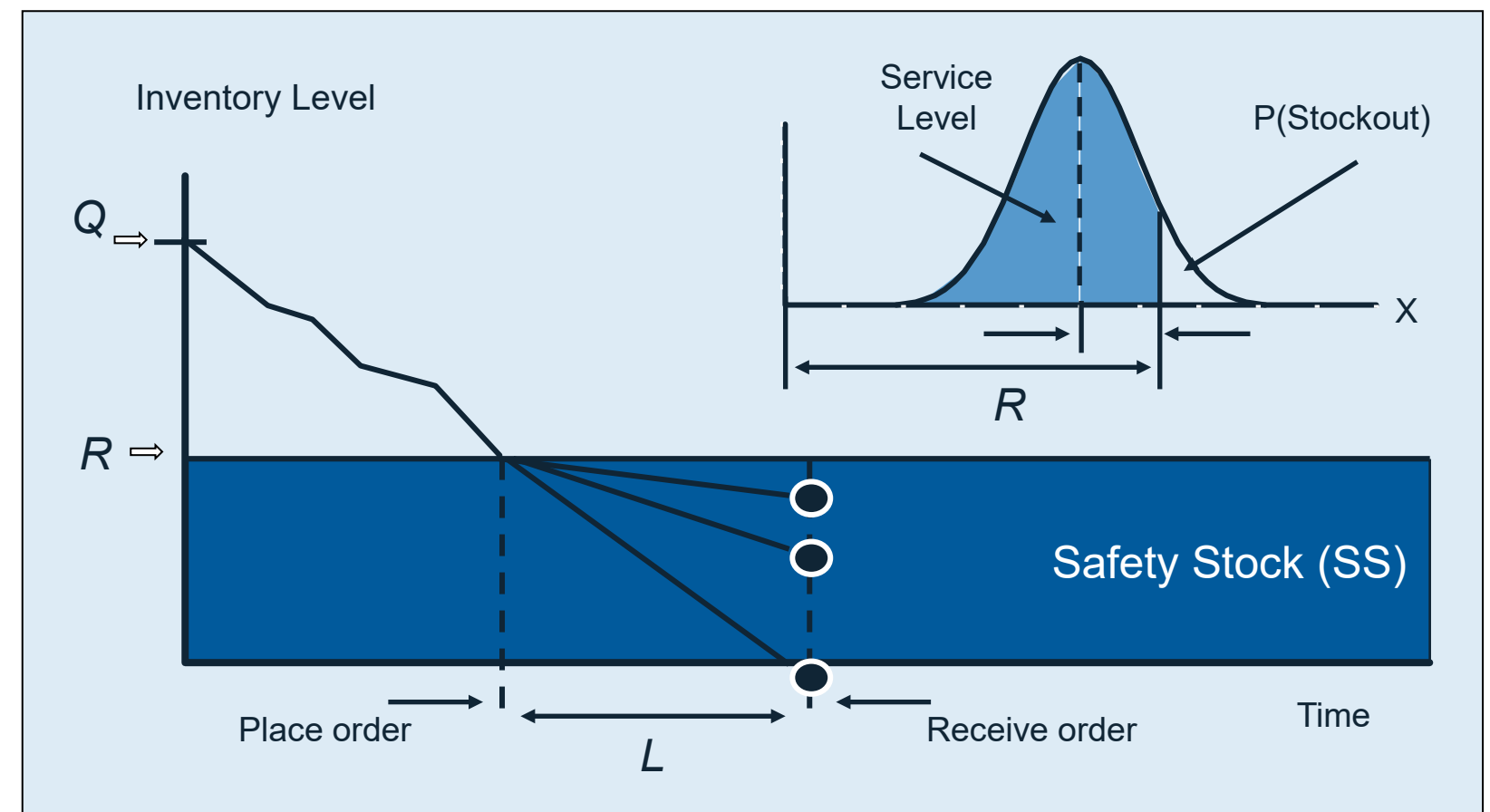
Basic Economic Order Quantity Model

- Basic economic order quantity model and reorder point behaviour can be analysed as follows:
- Intuitively: you put in your order when stocks start running low
- Define R as reorder point, and L delivery lead time
- Reorder point is $R = dL$, where d is daily demand



Probabilistic Models: When to Order?

- One can extend this formula to include a safety level that stock never runs out. A parameter z is used to include multiples of standard deviation. E.g. $z=1$ (1 x Standard deviation (σ)) yields a 16% probability retailer may run out of stock; $z=2$, or 2σ yields a 2.5%, $z=3$ or 3σ yields a 0.15% probability stocks may run out.
- The new re-order point is $R = dL + z\sigma$, where $z\sigma$ indicates the safety stock.



Inventory Management Trade-Offs

- Inventory helps deal with uncertainty and provides flexibility. For example when:
 - Supplies are delayed
 - Demand is greater than expected

Inventory Management Trade-Offs (cont.)

- However, inventory can be costly
 - Holding (or carrying) costs (interest payment on goods bought)
 - Costs for storage, handling, insurance, etc.
 - Costs for arranging specific equipment setups (relevant in manufacturing contexts)
 - Costs of canceling an order, etc.

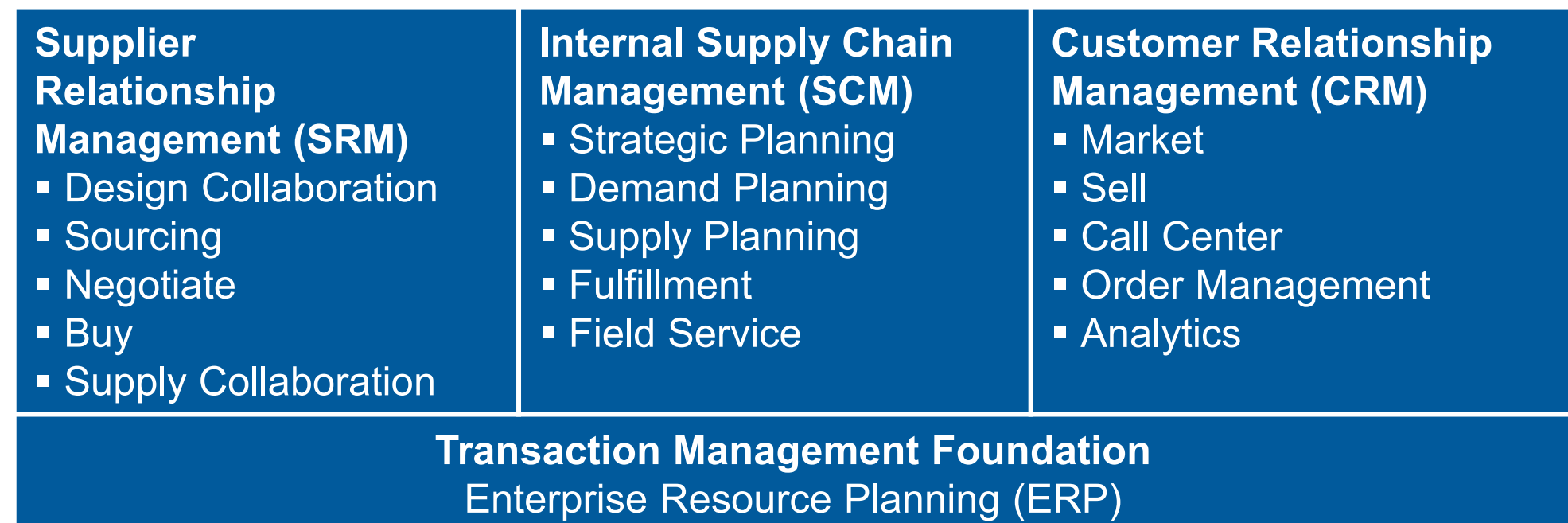
Approximate Inventory Holding Costs

- However, there is significant cost in keeping an inventory and its management
- Companies which minimise their inventory have significant cost advantage
- Here is a typical cost breakdown for inventories

<u>Category</u>	<u>Cost as a % of Inventory Value</u>
Housing costs (building rent, depreciation, operating cost, taxes, insurance)	6% (3 - 10%)
Material handling costs (equipment, lease or depreciation, power, operating cost)	3% (1 - 3.5%)
Labor cost from extra handling	3% (3 - 5%)
Investment costs (borrowing costs, taxes, and insurance on inventory)	11% (6 - 24%)
Pilferage, scrap, and obsolescence	3% (2 - 5%)
Overall carrying cost	26%

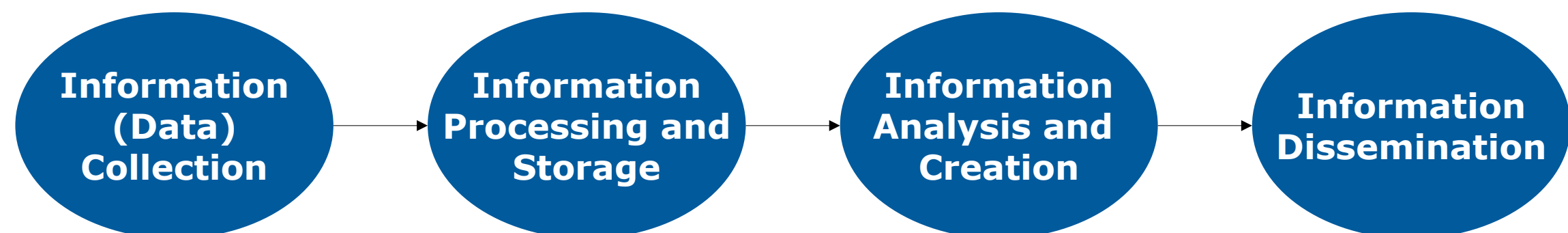
Supply Chain Management Infrastructure

- As supply chain management is a cross-functional approach to managing the movement of raw materials into an organization and the movement of finished goods out of the organization toward the end-consumer a number of applications/functions can be distinguished:



Information Flow

- Much of information flow (collection and dissemination) needs to be automated
 - Except for non-regular process information flow
- Database design needs to be harmonised across supply chain partners
- Much of data needs to be pre-processed before it is stored
- Supply chain functions are highly data dependent and therefore analysis can help make the supply process efficient



Summary

- Supply chains are important to all industries
- Smooth flow of goods and information can be the difference between good and bad businesses
- There exist many different variations of supply chain management
- Inventories help reduce risks
 - But they also cost
- Real-time information flow can reduce inventory costs significantly